PRINCE SAVSAVIYA

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Professional Summary

Machine Learning Engineer with strong expertise in **HPC and Distributed Systems**, specializing in building low-latency inference pipelines and optimizing GPU kernels. Experience deploying scalable AI solutions (Kubernetes/AWS) and accelerating deep learning workflows with CUDA/C++. Proven track record of improving search precision by 25% and reducing system latency by 40%.

Education

University of California, Riverside Riverside, CA March 2026 (expected)
M.S. Computer Science, GPA 3.56/4.0
Adani Institute of Infrastructure Engineering, GTU Ahmedabad, IN May 2024
B.E. Information & Communication Technology, GPA 3.6/4.0

Core Technical Skills

Machine Learning: PyTorch, TensorFlow, Scikit-learn, BERT, LLMs (Hugging Face), Computer Vision, Physics-Informed ML
MLOps & Cloud: Docker, Kubernetes, AWS (EKS, EC2, S3), GitHub Actions (CI/CD), Optuna, Vector Search, MLflow
Languages: Python, C++, SQL (PostgreSQL), Bash, CUDA, C
HPC & Performance: MPI, GPU Optimization, PETSc, DolfinX, Latency Optimization
Data & Search: Elasticsearch, Lucene, FAISS, Pandas, NumPy

Experience

Graduate Researcher — University of California, Riverside **Riverside, CA** **Jan 2025 – Present**

- Architected a modular **FSI simulation framework** orchestrating distributed CFD and FEM solvers via **MPI**, validated against Turek-Hron benchmarks for high-fidelity stability.
- Optimized **inter-process communication overhead** in partitioned solvers by refactoring data serialization pipelines, improving multi-node simulation throughput by **2**$\times$.
- Prototyping **Physics-Informed ML predictors** (Hamiltonian Neural Networks) to reduce fixed-point iterations; designing feature sets and data pipelines to accelerate model convergence.
- Developed a parameterized benchmark generator with automated regression tests (Python/Bash), reducing experiment setup time for new physics cases by **80%**.

Research Intern — Adani University **Ahmedabad, IN** **May 2023 – Dec 2023**

- Built a physics-informed ML framework combining PDE ageing models with ensemble regressors (SVR, RF, GBR), **cutting MSE 98%** and **RMSE 86%** vs. data-only baseline.
- Ingested **200,000+** battery-cycle records in Python; automated feature engineering and Optuna hyperparameter tuning, reducing experiment turnaround **30%**.
- **Publication:** *Journal of Power Sources*, Vol. 627, 2025 (IF $\approx$ 7.9) — DOI 10.1016/j.jpowsour.2024.235771.

Selected Projects

RetrieveX — Semantic Searchbo Engine **Django · BERT · FAISS · AWS** **Jan 2025 – Apr 2025**

- Architected a hybrid retrieval system merging BM25 keyword search with **BERT+FAISS vector embeddings**, improving top-3 precision **25%** on 150k medical documents.
- Engineered an asynchronous inference pipeline using Django and REST APIs, optimizing query latency to $<$**200 ms** (P_{95}).

AI-Driven Infrastructure Autoscaler **LSTM · Kubernetes · Helm · CI/CD** **Jan 2025 – Apr 2025**

- Designed a time-series forecasting service using LSTMs to predict cluster load, automating replica scaling and cutting cloud over-provisioning costs by **18%**.
- Implemented MLOps best practices with automated model retraining triggers and CI/CD deployment to AWS EKS; sustained 99.9% SLA under $4\times$ load spikes.

Risk-Analytics Microservice **FastAPI · Pandas · Docker · AWS** **Sep 2024 – Jan 2025**

- Designed real-time analytics service ingesting multi-asset data with vectorized pipelines; delivered metrics in $<$**150 ms**.
- Automated deployment with GitHub Actions $\rightarrow$ AWS ECR $\rightarrow$ EC2; maintained 99.9% uptime with Prometheus monitoring.